HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and produced **2D and 3D engineering drawings** for novel machinery components and complex fabricated assemblies using **SolidWorks**, adhering to strict manufacturing tolerances.
- Assisted senior engineers in the **design and development** of industrial augers and screw conveyors, translating conceptual designs into detailed manufacturing specifications.
- Prepared precise **sheet metal fabrication drawings**, including bend allowances, weld symbols, and material callouts, ensuring clarity for the workshop floor.
- Collaborated with the **production team** to review and refine designs, incorporating feedback on manufacturability and material optimisation for fabricated parts.
- Modified and updated **existing engineering drawings** based on product revisions and customer feedback, maintaining an accurate revision history within the drawing management system.
- Supported the generation of **material specifications and bills of materials** for new product designs, ensuring accurate component sourcing and cost tracking.
- Contributed to ongoing **product development and engineering projects**, providing technical documentation and design support across multiple concurrent initiatives.

SKILLS

CAD Software – SolidWorks, AutoCAD

Engineering Disciplines – Mechanical Engineering, Engineering Design, Product Design, Manufacturing Processes, Sheet Metal Fabrication, Welding, Workshop Processes

Core Competencies – 2D Drawing, 3D Modelling, Bills of Materials, Material Specifications, Design Development, Technical Documentation, Problem Solving

Machinery & Fabrication – Industrial Augers, Screw Conveyors, Tub Feeders, Associated Machinery, Fabricated Components

Industry Exposure – Agriculture, Food, Pharmaceutical, Construction, Waste Management, Recycling, Manufacturing Environments

Soft Skills – Communication, Teamwork, Attention to Detail, Practical Approach, Hands-on Ability, Willingness to Learn

PROJECTS

Auger Flighting Design Optimisation (HCL Technologies)

- Engineered detailed **3D models and fabrication drawings** for custom auger flighting profiles using **SolidWorks**, enhancing material flow efficiency for an industrial client.
- Performed kinematic simulations to validate screw conveyor movement and clearance, identifying potential interference points and refining designs for smoother operation.
- Generated comprehensive **bills of materials and material specifications** for fabricated components, coordinating with the procurement team to ensure timely delivery of raw materials.

Conveyor System Component Design (HCL Technologies)

- Created precise **2D manufacturing drawings** for conveyor support structures and drive mechanisms, incorporating specific tolerances for welding and assembly.
- Assisted in the selection of appropriate materials and fasteners, documenting specifications to meet operational load and environmental requirements.
- Collaborated with the HCL production liaison to ensure designs were practical for fabrication and assembly within the company's workshop capabilities.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)